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1. Implement a hash table using Linear Probing

Program:

```
#include<stdio.h>

#define SIZE 10

int main(){
    int hash[SIZE],n,key,i,index;
    for(i=0;i<SIZE;i++)
        hash[i]=-1;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements:\n");
    for(i=0;i<n;i++){
        scanf("%d",&key);
        index=key%SIZE;
        while(hash[index]!=-1)
            index=(index+1)%SIZE;
        hash[index]=key;
    }
    printf("Hash Table:\n");
    for(i=0;i<SIZE;i++)
        printf("%d : %d\n",i,hash[i]);
    return 0;
}
```

Terminal Process:

```
PS C:\Users\chand\OneDrive\DAA-2026> gcc hashline.c -o has
PS C:\Users\chand\OneDrive\DAA-2026> ./has
Enter number of elements: 5
Enter elements:
23
34
53
13
12
Hash Table:
0 : -1
1 : -1
2 : 12
3 : 23
4 : 34
5 : 53
6 : 13
7 : -1
8 : -1
9 : -1
PS C:\Users\chand\OneDrive\DAA-2026> |
```

2. Implement a hash table using Quadratic Probing

Program:

```
#include<stdio.h>

#define SIZE 10

int main(){
    int hash[SIZE],n,key,i,index,j;

    for(i=0;i<SIZE;i++)
        hash[i]=-1;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++){
        scanf("%d",&key);
```

```

    index=key%SIZE;
    j=0;
    while(hash[(index+j*j)%SIZE]!=-1)
        j++;
    hash[(index+j*j)%SIZE]=key;
}
printf("Hash Table:\n");
for(i=0;i<SIZE;i++)
    printf("%d : %d\n",i,hash[i]);
return 0;
}

```

Terminal Process:

```

● PS C:\Users\chand\OneDrive\DAA-2026> gcc hashquad.c -o has
● PS C:\Users\chand\OneDrive\DAA-2026> ./has
Enter number of elements: 7
Enter elements:
12
32
34
54
65
67
78
Hash Table:
0 : -1
1 : -1
2 : 12
3 : 32
4 : 34
5 : 54
6 : 65
7 : 67
8 : 78
9 : -1
○ PS C:\Users\chand\OneDrive\DAA-2026>

```

3. Given an array and a target value T, determine whether there are two elements whose sum is equal to T. (Average Time Complexity should be $O(n)$)

Program:

```
#include<stdio.h>

#define SIZE 100

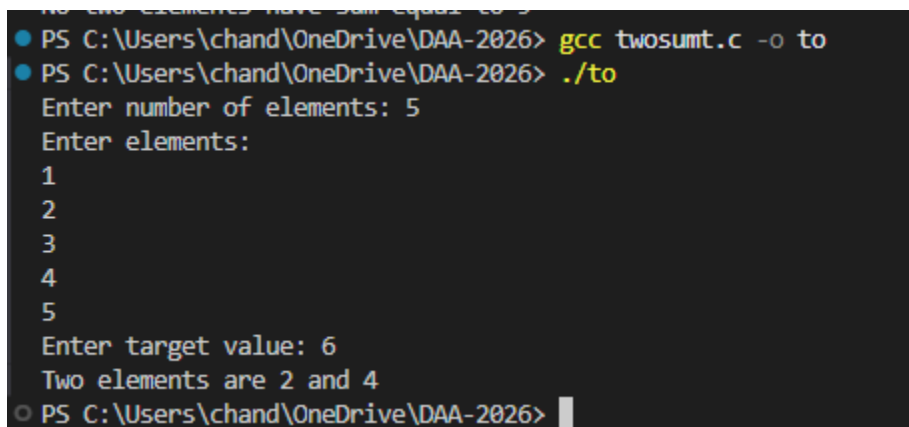
int main(){
    int a[SIZE],hash[SIZE],n,T,i,index,j,found=0;
    for(i=0;i<SIZE;i++)
        hash[i]=-1;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements:\n");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
        printf("Enter target value: ");
        scanf("%d",&T);
        for(i=0;i<n;i++){
            index=(T-a[i])%SIZE;
            if(index<0)
                index+=SIZE;
            j=0;
            while(hash[(index+j)%SIZE]!=-1){
                if(hash[(index+j)%SIZE]==T-a[i]){
                    found=1;
                    break;
                }
            }
        }
    }
}
```

```

    }
    j++;
}
if(found){
    printf("Two elements are %d and %d\n",T-a[i],a[i]);
    break;
}
index=a[i]%SIZE;
if(index<0)
    index+=SIZE;
j=0;
while(hash[(index+j)%SIZE]!=-1)
    j++;
hash[(index+j)%SIZE]=a[i];
}
if(!found)
    printf("No two elements have sum equal to %d\n",T);
return 0;
}

```

Terminal Process:



```

PS C:\Users\chand\OneDrive\DAA-2026> gcc twosumt.c -o to
PS C:\Users\chand\OneDrive\DAA-2026> ./to
Enter number of elements: 5
Enter elements:
1
2
3
4
5
Enter target value: 6
Two elements are 2 and 4
PS C:\Users\chand\OneDrive\DAA-2026>

```

4. Given an integer array, use hashing to find the frequency of every distinct element.

Program:

```
#include<stdio.h>

#define SIZE 100

int main(){
    int a[SIZE],hash[SIZE],freq[SIZE],n,i,key,index,j;
    for(i=0;i<SIZE;i++){
        hash[i]=-1;
        freq[i]=0;
    }
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements:\n");
    for(i=0;i<n;i++){
        scanf("%d",&key);
        index=key%SIZE;
        if(index<0)
            index+=SIZE;
        j=0;
        while(hash[(index+j)%SIZE]!=-1&&hash[(index+j)%SIZE]!=key)
            j++;
        if(hash[(index+j)%SIZE]==-1)
            hash[(index+j)%SIZE]=key;
        freq[(index+j)%SIZE]++;
    }
}
```

```

    }

    printf("Frequency of elements:\n");
    for(i=0;i<SIZE;i++)
        if(hash[i]!=-1)
            printf("%d : %d\n",hash[i],freq[i]);
    return 0;
}

```

Terminal Process:

```

● PS C:\Users\chand\OneDrive\DAA-2026> gcc freq.c -o fre
● PS C:\Users\chand\OneDrive\DAA-2026> ./fre
Enter number of elements: 8
Enter elements:
1
3
67
6
3
3
68
98
Frequency of elements:
1 : 1
3 : 3
6 : 1
67 : 1
68 : 1
98 : 1
○ PS C:\Users\chand\OneDrive\DAA-2026> 

```